Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B. Tech | *Time: 03 Hours*

Code: 102611 | *Subject: Renewable Energy Systems* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Renewable energy is best defined as
- (i) Energy stored in fossil fuels
- (ii) Energy obtained from continuously replenished sources
- (iii) Energy from nuclear fission
- (iv) Energy from biomass only

(b) One major scientific principle behind renewable energy is
- (i) Conservation of momentum
- (ii) Conversion of natural flows into usable energy
- (iii) Use of chemical reactions only
- (iv) Elimination of entropy

(c) A key technical implication of renewable energy adoption is
- (i) Zero initial investment
- (ii) Unlimited storage
- (iii) Intermittency of energy sources like solar and wind
- (iv) Reduced need for power electronics

(d) Solar collector geometry primarily affects
- (i) Orientation and tilt for maximum solar gain
- (ii) Cloud formation
- (iii) Air density
- (iv) Earth's rotation

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(e) A common application of solar cells is
- (i) Chemical distillation
- (ii) Producing hydrogen by electrolysis directly
- (iii) Solar pond heating
- (iv) Electricity generation for remote areas

(f) An impulse turbine is best suited for
- (i) Low head and high flow
- (ii) High head and low flow
- (iii) Underground reservoirs
- (iv) Tidal fluctuations

(g) Pyrolysis involves
- (i) Burning biomass with excess air
- (ii) Mixing ethanol with gasoline
- (iii) Thermal decomposition of biomass in absence of oxygen
- (iv) Compressing methane

(h) The power extracted by a wind turbine mainly depends on
- (i) Turbine height only
- (ii) Air density, rotor area, and wind velocity
- (iii) Hub material
- (iv) Number of tower supports

(i) Anaerobic digestion produces
- (i) Hydrogen
- (ii) Biogas (mainly methane)
- (iii) Oxygen
- (iv) Ammonia

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(j) Blade Element Theory is primarily used to

(i) Determine aerodynamic forces along the blade span
(ii) Estimate thermal stresses in blades
(iii) Measure blade vibration
(iv) Design gearboxes

Q2 (a) Explain the impact of renewable energy generation on environment in detail. [7]
(b) Summarize about the various types of non-conventional energy sources [7]

Q3 (a) Explain the in detail about the solar radiation phenomena. [7]

(b) Define solar irradiance, solar constant, extraterrestrial and terrestrial radiations. [7]

Q.4 (a) Summarize in detail about the construction and working principle of Solar Thermal Power Plant. [7]

(b) Explain in detail about the construction of solar cell, and solar module [7]

Q.5 (a) Explain the in detail about the solar radiation phenomena. [7]
(b) With a neat sketch, explain the principle of multi-effect solar desalination using parabolic trough concentrators. [7]

Q.6 (a) Discuss the advantages and environmental concerns of solar pond-based desalination. [7]
(b) What is the meaning of biomass? Further, discuss its multipurpose utilization [7]

Q.7 (a) Differentiate between the following methods of biogas generation [7]
i. Pyrolysis ii. Combustion
(b) With a neat sketch explain the operation impoundment hydro power plant [7]

Q.8 (a) Explain the operation of the following Reaction turbines: i. Francis ii. Kaplan [7]
(b) Discuss about selection of water turbine based on capacity of the power plan, head and water flow rate. [7]

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Q.9 (a) With a neat diagram, explain how wind energy can be converted into electrical energy [7]
(b) Distinguish the difference between vertical axis wind turbine and horizontal axis wind turbine. [7]

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